

Order of subshell filling

The sequence in which the subshells are filled is given in most cases by the Aufbau principle, also known as the Madelung or Klechkovsky rule (after Erwin Madelung and Vsevolod Klechkovsky respectively). This rule was first observed empirically by Madelung, and Klechkovsky and later authors gave it theoretical justification.^{[46][47][48][49][g]} The shells overlap in energies, and the Madelung rule specifies the sequence of filling according to:^[47]

$$1s \ll 2s < 2p \ll 3s < 3p \ll 4s < 3d < 4p \ll 5s < 4d < 5p \ll 6s < 4f < 5d < 6p \ll 7s < 5f < 6d < 7p \ll \dots$$

Here the sign $\ll$ means "much less than" as opposed to $<$ meaning just "less than".^[47] Phrased differently, electrons enter orbitals in order of increasing $n + \ell$, and if two orbitals are available with the same value of $n + \ell$, the one with lower n is occupied first.^{[45][49]} In general, orbitals with the same value of $n + \ell$ are similar in energy, but in the case of the s orbitals (with $\ell = 0$), quantum effects raise their energy to approach that of the next $n + \ell$ group. Hence the periodic table is usually drawn to begin each row (often called a period) with the filling of a new s orbital, which corresponds to the beginning of a new shell.^{[47][48][30]} Thus, with the exception of the first row, each period length appears twice.^[47]

2, 8, 8, 18, 18, 32, 32, ...

The overlaps get quite close at the point where the d orbitals enter the picture,^[50] and the order can shift slightly with atomic number^[51] and atomic charge.^{[52][h]}

Starting from the simplest atom, this lets us build up the periodic table one at a time in order of atomic number, by considering the cases of single atoms. In hydrogen, there is only one electron, which must go in the lowest-energy orbital 1s. This electron configuration is written $1s^1$, where the superscript indicates the number of electrons in the subshell. Helium adds a second electron, which also goes into 1s, completely filling the first shell and giving the configuration $1s^2$.^{[39][58][i]}

Starting from the third element, lithium, the first shell is full, so its third electron occupies a 2s orbital, giving a $1s^2 2s^1$ configuration. The 2s electron is lithium's only valence electron, as the 1s subshell is now too tightly bound to the nucleus to participate in chemical bonding to other atoms: such a shell is called a "core shell". The 1s subshell is a core shell for all elements from lithium onward. The 2s subshell is completed by the next element beryllium ($1s^2 2s^2$). The following elements then proceed to fill the 2p subshell. Boron ($1s^2 2s^2 2p^1$) puts its new electron in a 2p orbital; carbon ($1s^2 2s^2 2p^2$) fills a second 2p orbital; and with nitrogen ($1s^2 2s^2 2p^3$) all three 2p orbitals become singly occupied. This is consistent with Hund's rule, which states that atoms usually prefer to singly occupy each orbital of the same type before filling them with the second electron. Oxygen ($1s^2 2s^2 2p^4$), fluorine ($1s^2 2s^2 2p^5$), and neon ($1s^2 2s^2 2p^6$) then complete the already singly filled 2p orbitals; the last of these fills the second shell completely.^{[39][58]}

Starting from element 11, sodium, the second shell is full, making the second shell a core shell for this and all heavier elements. The eleventh electron begins the filling of the third shell by occupying a 3s orbital, giving a configuration of $1s^2 2s^2 2p^6 3s^1$ for sodium. This configuration is abbreviated [Ne] $3s^1$, where [Ne] represents neon's configuration. Magnesium ([Ne] $3s^2$) finishes this 3s orbital, and the following six elements aluminium, silicon, phosphorus, sulfur, chlorine, and argon fill the three 3p orbitals ([Ne] $3s^2 3p^1$ through [Ne] $3s^2 3p^6$).^{[39][58]} This creates an analogous series in which the outer shell structures of sodium through argon are analogous to those of lithium through neon, and is the basis for the periodicity of chemical properties that the periodic table illustrates.^[39] at regular but changing intervals of atomic numbers, the properties of the chemical elements approximately repeat.^[36]

The first 18 elements can thus be arranged as the start of a periodic table. Elements in the same column have the same number of valence electrons and have analogous valence electron configurations: these columns are called groups. The single exception is helium, which has two valence electrons like beryllium and magnesium, but is typically placed in the column of neon and argon to emphasise that its outer shell is full. (Some contemporary authors question even this single exception, preferring to consistently follow the valence configurations and place helium over beryllium.) There are eight columns in this periodic table fragment, corresponding to at most eight outer-shell electrons.^[32] A period begins when a new shell starts filling.^[30] Finally, the colouring illustrates the blocks: the elements in the s-block (coloured red) are filling s orbitals, while those in the p-block (coloured yellow) are filling p orbitals.^[30]

1 H							2 He	2×1 = 2 elements 1s
3 Li	4 Be	5 B	6 C	7 N	8 O	9 F	10 Ne	2×(1+3) = 8 elements 2s 2p
11 Na	12 Mg	13 Al	14 Si	15 P	16 S	17 Cl	18 Ar	2×(1+3) = 8 elements 3s 3p

Starting the next row, for potassium and calcium the 4s subshell is the lowest in energy, and therefore it fills next.^{[39][58]} Potassium adds one electron to the 4s shell ([Ar] $4s^1$), and calcium then completes it ([Ar] $4s^2$). However, starting from scandium ([Ar] $3d^1 4s^2$) the 3d subshell becomes the next highest in energy. The 4s and 3d subshells have approximately the same energy and they compete for filling the electrons, and so the occupation is not quite consistently filling the 3d orbitals one at a time. The precise energy ordering of 3d and 4s changes along the row, and also changes depending on how many electrons are removed from the atom. For example, due to the repulsion between the 3d electrons and the 4s ones, at chromium the 4s energy level becomes slightly higher than 3d, and so it becomes more profitable for a chromium atom to have a [Ar] $3d^5 4s^1$ configuration than an [Ar] $3d^4 4s^2$ one. A similar anomaly occurs at copper, whose atom has a [Ar] $3d^{10} 4s^1$ configuration rather than the expected [Ar] $3d^9 4s^2$.^[39] These are violations of the Madelung rule. Such anomalies, however, do not have any chemical significance:^[52] most chemistry is not about isolated gaseous atoms,^[60] and the various configurations are so close in energy to each other^[50] that the presence of a nearby atom can shift the balance.^[39] Therefore, the periodic table ignores them and considers only idealized configurations.^[38]

At zinc ([Ar] $3d^{10} 4s^2$), the 3d orbitals are completely filled with a total of ten electrons.^{[39][58]} Next come the 4p orbitals, completing the row, which are filled progressively by gallium ([Ar] $3d^{10} 4s^2 4p^1$) through krypton ([Ar] $3d^{10} 4s^2 4p^6$), in a manner analogous to the previous p-block elements.^{[39][58]} From gallium onwards, the 3d orbitals form part of the electronic core, and no longer participate in chemistry.^[57] The s- and p-block elements, which fill their outer shells, are called main-group elements; the d-block elements (coloured blue below), which fill an inner shell, are called transition elements (or transition metals, since they are all metals).^[61]

